PS 2010: 11. Ordinary Least Squares(OLS)

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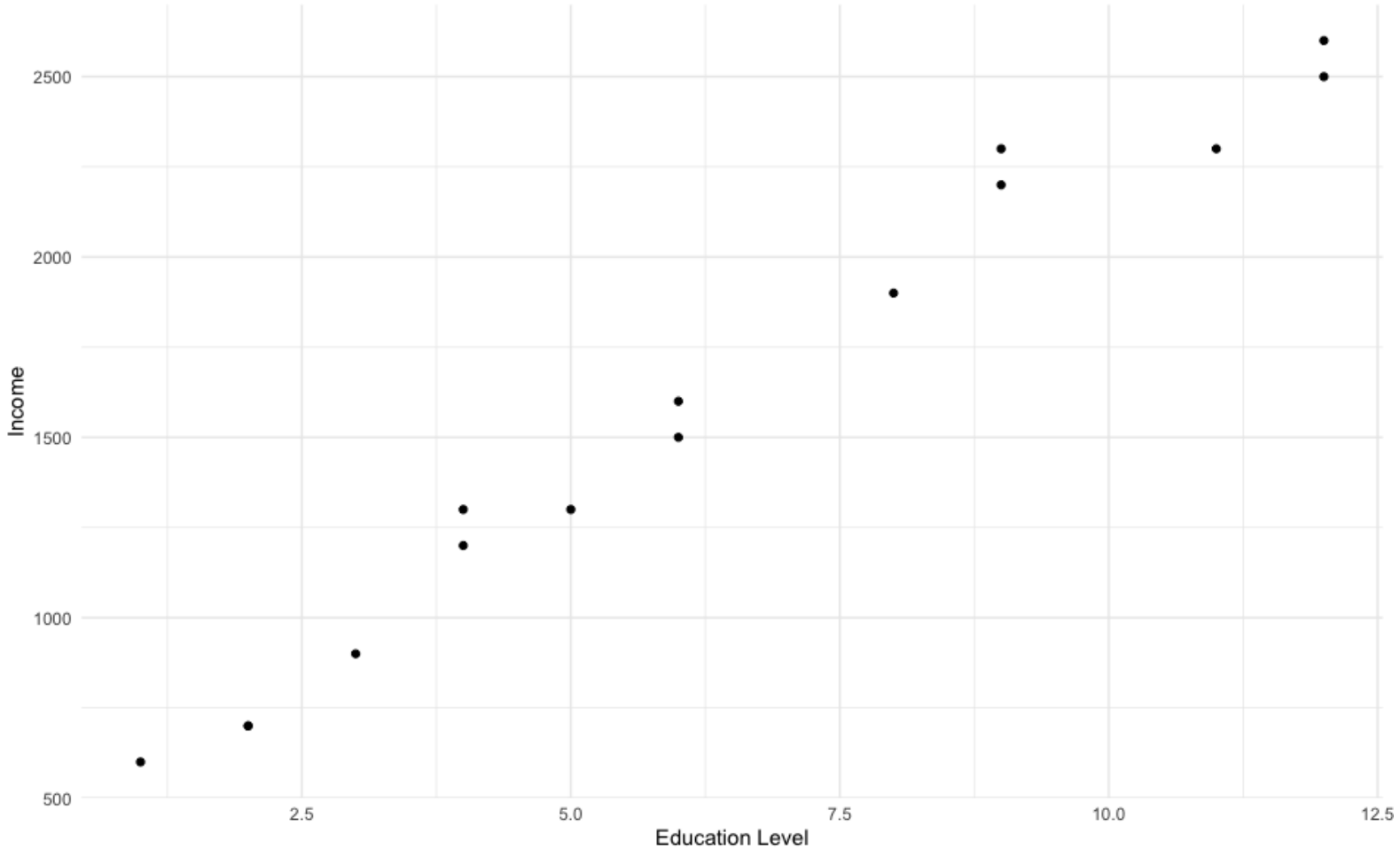
- Homework 6 due today
- No class next Monday
- No homework but today's class is very important

Agenda

- OLS estimator
- Hypothesis Testing

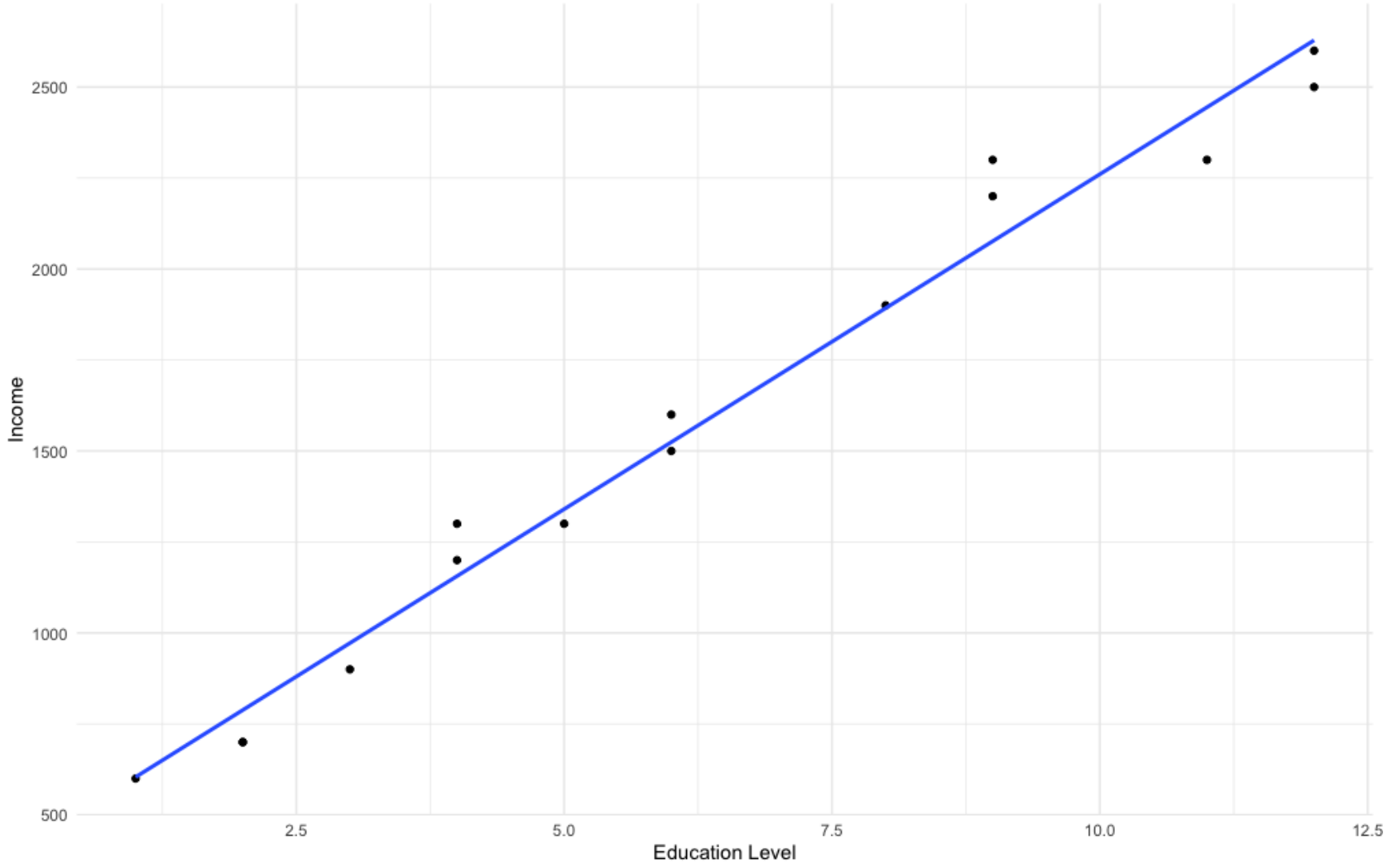
A Linear Model

Suppose we have data on education level and income and want to know their relationship



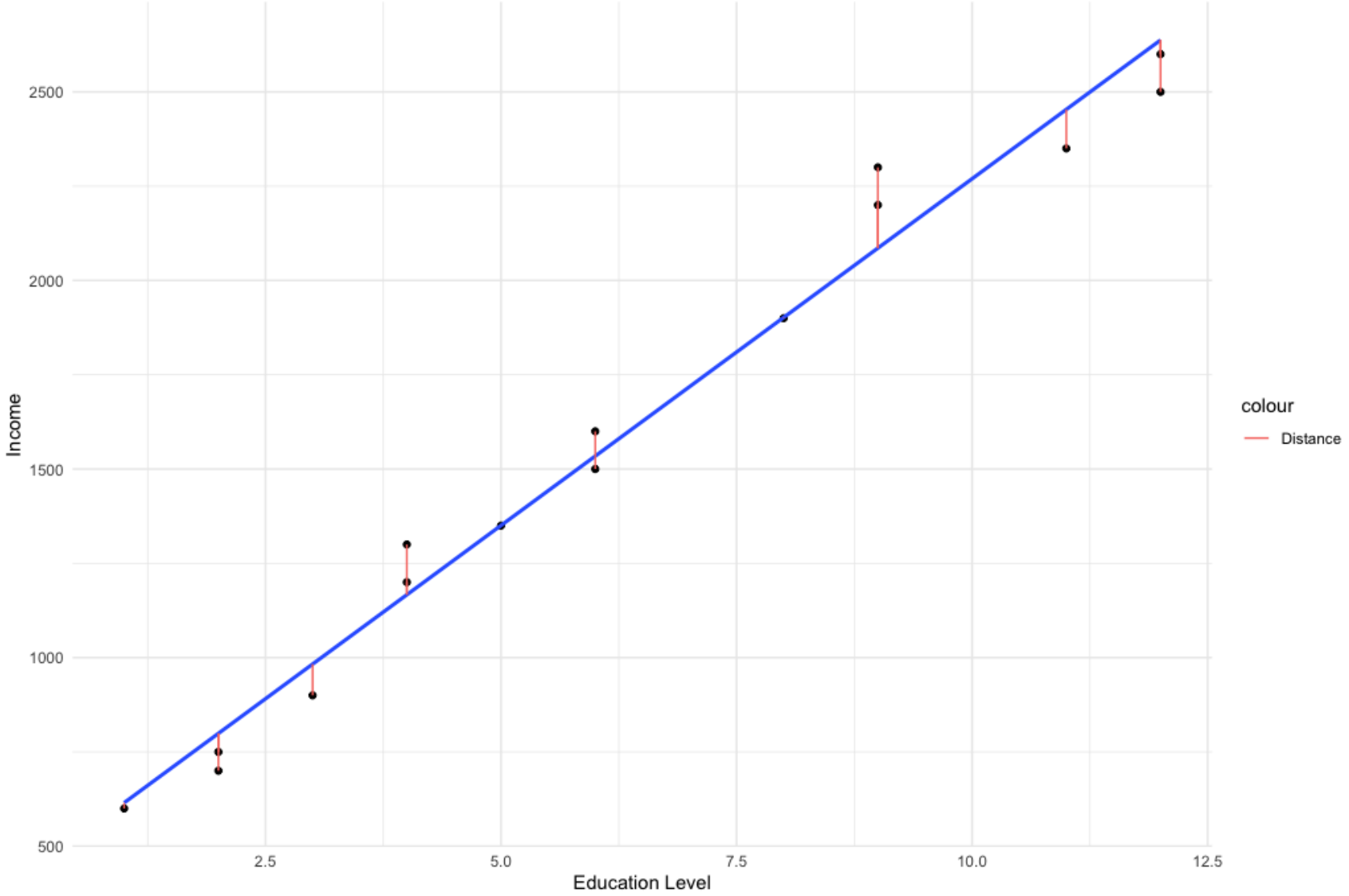
A Linear Model

The goal is to find a line that best describe the linear relationship



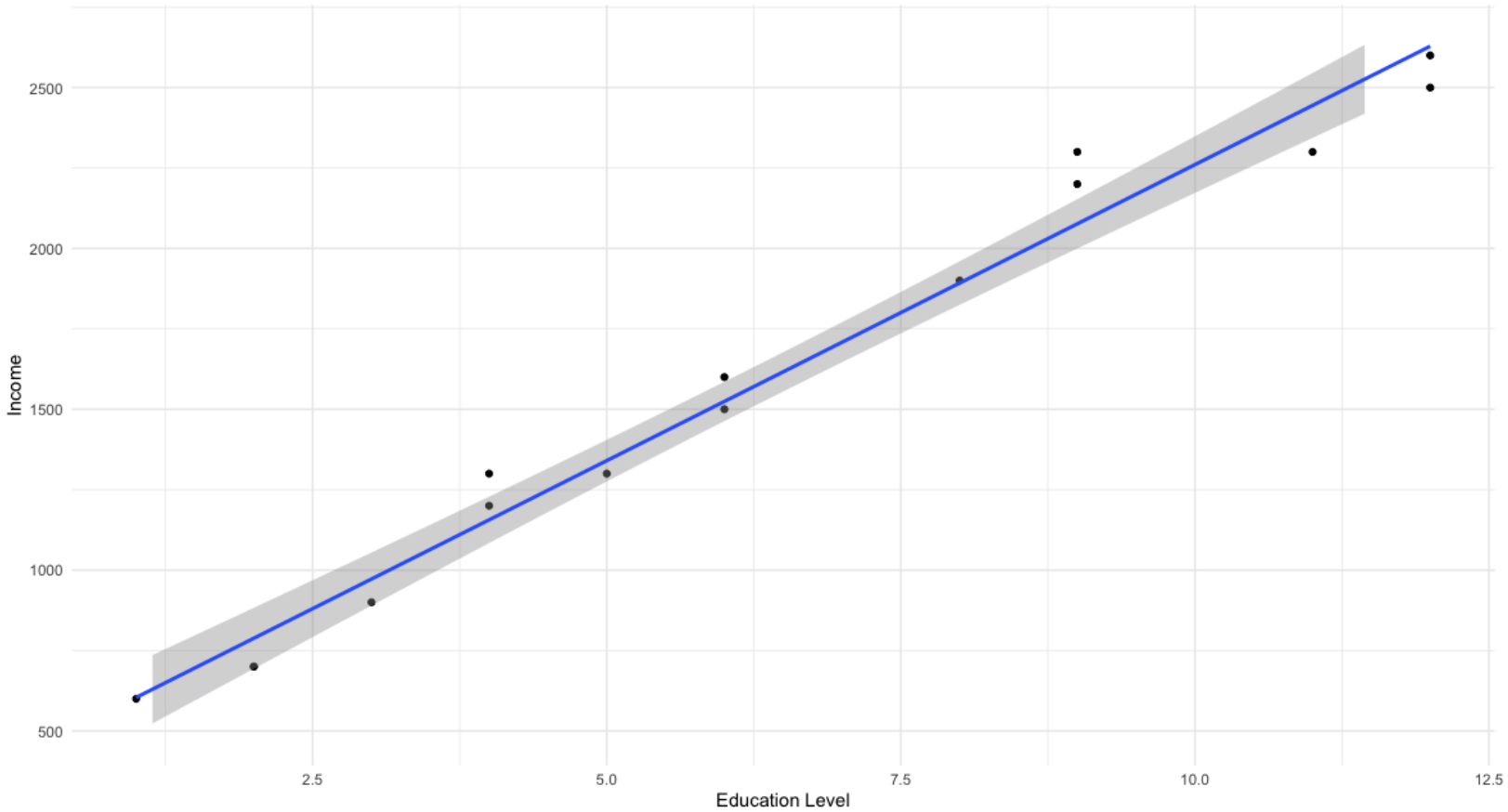
A Linear Model

We minimize the distance between line and data points



A Linear Model

We also want to describe the uncertainty



OLS

Let's formalize our idea a little bit

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ \vdots \\ Y_n \end{bmatrix}_{n\times 1} = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix}_{n\times k} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \vdots \\ \beta_k \end{bmatrix}_{k\times 1} + \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ \vdots \\ e_n \end{bmatrix}_{n\times 1}$$

Which can be simplified as

$$y = X\beta + e$$

Our job is to find $\hat{\beta}$

$$y = X\hat{\beta} + \epsilon$$

OLS

- Our goal is to minimize sum of distance
- One way is to minimize sum of squared distance
- That is $\epsilon^{'}\epsilon$
- We have $\epsilon = y - X\hat{\beta}$

$$\begin{bmatrix} \epsilon_1 & \epsilon_2 & \dots & \epsilon_n \end{bmatrix}_{1\times n} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}_{n\times 1} = [\epsilon_1 \times \epsilon_1 + \epsilon_2 \times \epsilon_2 + \ldots + \epsilon_n \times \epsilon_n]_{1\times 1}$$

OLS

$$
\begin{aligned}
\epsilon'\epsilon &= (y - X\hat{\beta})'(y - X\hat{\beta}) \\
&= y'y - \hat{\beta}'X'y - y'X\hat{\beta} + \hat{\beta}'X'X\hat{\beta} \\
&= y'y - 2\hat{\beta}'X'y + \hat{\beta}'X'X\hat{\beta}
\end{aligned}
$$

Where $\hat{\beta}'X'y$ and $y'X\hat{\beta}$ are scalars

OLS

$$\epsilon'\epsilon = y'y - 2\hat{\beta}'X'y + \hat{\beta}'X'X\hat{\beta}$$

So, the goal is to find $\hat{\beta}$ that minimize the above equation

$$\frac{\partial \epsilon'\epsilon}{\partial \hat{\beta}} = -2X'y + 2X'X\hat{\beta} = 0$$

Then

$$X'X\hat{\beta} = X'y$$

Finally

$$\hat{\beta} = (X'X)^{-1}X'y$$

Assumptions

- We make no assumptions to derive OLS estimator
- However, we need make some assumptions to guarantee OLS is BLUE
- And allow us to do inference

Gauss-Markov Assumptions

- **Assumption 1:** Linear relationship between dependent and independent variables

$$y = X\beta + e$$

- **Assumption 2:** No perfect multicollinearity in independent variables

$$rank(X) = k$$

- **Assumption 3:** Strict exogeneity or zero conditional mean assumption

$$E(e \mid X) = 0$$

- **Assumption 4:** Spherical error variance
  - **Homoskedasticity:** $E(e'e \mid X) = \sigma^2$
  - **No autocorrelation:** $E(e_i e_j \mid X) = 0$ for i, j = 1,2,3,...; $i \neq j$

Unbiaseness

- The Gauss-Markov Assumptions guarantee OLS estimator is BLUE
- Unbiasedness:

$$
\begin{aligned}
\hat{\beta} &= (X'X)^{-1}X'y \\
\hat{\beta} &= (X'X)^{-1}X'(X\beta + e) \\
\hat{\beta} &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'e \\
\hat{\beta} &= \beta + (X'X)^{-1}X'e \\
E[\hat{\beta}] &= E[\beta + (X'X)^{-1}X'e] \\
&= \beta + (X'X)^{-1}X'E[e]
\end{aligned}
$$

- $E[e] = 0$ by assumption
- Therefore, we have $E[\hat{\beta}] = \beta$

Variance-covariance Structure

What is variance-covariance structure for $\hat{\beta}$ looks like, given Gauss-Markov Assumptions

$$\begin{aligned}\hat{\beta} &= \beta + (X'X)^{-1}X'e \\ V[\hat{\beta}] &= V[(X'X)^{-1}X'e] \\ &= (X'X)^{-1}X'V[e]X'(X'X)^{-1}\end{aligned}$$

We have $V[e] = E(e'e) - [E(e)]^2 = E(e'e) = \sigma^2$

$$\begin{aligned}V[\hat{\beta}] &= (X'X)^{-1}X'E(e'e)X'(X'X)^{-1} \\ &= (X'X)^{-1}X'(\sigma^2 I)X'(X'X)^{-1} \\ &= \sigma^2(X'X)^{-1}X'X'(X'X)^{-1} \\ &= \sigma^2(X'X)^{-1}\end{aligned}$$

Variance-covariance

What does the variance-covariance matrix of the OLS estimator look like?

Remember that error variance looks like this

$$\sigma^2 I = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix}$$

Therefore, the variance-covariance matrix for the OLS estimator:

$$V[\hat{\beta}] = \begin{bmatrix} \text{var}\left(\hat{\beta}_1\right) & \text{cov}\left(\hat{\beta}_1, \hat{\beta}_2\right) & \dots & \text{cov}\left(\hat{\beta}_1, \hat{\beta}_k\right) \\ \text{cov}\left(\hat{\beta}_2, \hat{\beta}_1\right) & \text{var}\left(\hat{\beta}_2\right) & \dots & \text{cov}\left(\hat{\beta}_2, \hat{\beta}_k\right) \\ \vdots & \vdots & \vdots & \vdots \\ \text{cov}\left(\hat{\beta}_k, \hat{\beta}_1\right) & \text{cov}\left(\hat{\beta}_k, \hat{\beta}_2\right) & \dots & \text{var}\left(\hat{\beta}_k\right) \end{bmatrix}$$

Non-Spherical Errors (Not Important!)

- Heteroskedasticity but no correlation between observations

$$\sigma^2 I = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{bmatrix}$$

- Heteroskedasticity and correlation between observations

$$\Omega = \begin{bmatrix} \Omega_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \Omega_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \Omega_G \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \cdot & \cdot & 0 & 0 & 0 \\ \sigma_{12} & \sigma_2^2 & \sigma_{23} & \cdot & \cdot & 0 & 0 & 0 \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 & \cdot & \cdot & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot & \sigma_1^2 & \sigma_{12} & \sigma_{13} \\ 0 & 0 & 0 & \cdot & \cdot & \sigma_{12} & \sigma_2^2 & \sigma_{23} \\ 0 & 0 & 0 & \cdot & \cdot & \sigma_{13} & \sigma_{23} & \sigma_3^2 \end{bmatrix}$$

Hypothesis Testing

- We now have our estimates and their corresponding variance
- we can conduct single or joint hypothesis testing
- To calculate statistic, we need one more assumption for error term

$$e \sim N\left[0, \sigma^2 I\right]$$

- For single parameter testing

$$H_0 : \hat{\beta}_k = c$$
$$H_1 : \hat{\beta}_k \neq c$$

$$T = \frac{\hat{\beta}_k - c}{\sqrt{v(\hat{\beta}_{kk})}}$$

Hypothesis Testing

- If sample is small then it is t, if large, then z
- The important point is: we need error variance σ^2 to calculate variance matrix for $\hat{\beta}$. However, we do not know it

$$V(\hat{\beta}) = \sigma^2 (X'X)^{-1}$$

- So, we use estimated error variance to replace

$$\hat{\sigma}^2 = \frac{(Y - \mathbf{X}\hat{\beta})'(Y - \mathbf{X}\hat{\beta})}{n-k} = \frac{\epsilon'\epsilon}{n-k}$$

- We also need calculate $t_{\alpha/2,n-k}$ for critical value
- Notice, we have k equal to the total parameters we estimate: independent variable + 1

Confidence Interval

- Because we have $SE(\beta_k)$, we can also construct confidence interval

$$\mathrm{CI}_{1-\alpha}(\hat{\beta}_k) = [\hat{\beta}_k - z_\alpha \,\mathrm{SE}(\hat{\beta}_k), \hat{\beta}_k + z_\alpha \,\mathrm{SE}(\hat{\beta}_k)]$$

- Change z to t in small sample

R-square

- How do you know if your model is a good model?
- How many proportion of variance in dependent variable is explained by your model?
- Ok, this means we need to know two things:
  - Total variance in dependent variable
  - Variance explained by our model
- Total variance $= \sum_i^n (y_i - \bar{y})^2$
- % of Explained variance = 1 - % of unexplained variance
- We know unexplained variance$= \sum_i^n \epsilon_i^2$

$$R^2 = 1 - \frac{\sum_i^n \epsilon_i^2}{\sum_i^n (y_i - \bar{y})^2}; R^2_{Adjust} = 1 - (1 - R^2) * \frac{n-1}{n-k}$$

Joint Hypotheses Testing

$$H_0 : \hat{\beta}_1 = \hat{\beta}_2 = ...\hat{\beta}_k = 0$$

$$H_1 : \textit{One of them or All of them are non zero}$$

$$F = \frac{\text{Explained variance}/(k-1)}{\text{Unexplained variance}/(n-k)}$$

The critical value for significant level α is F With k-1 and n-k degree of freedom